



Continuum Mechanics

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Chapter 02

Tensor Calculus

Part 4





Definition (2.20)

A function

$$\psi : \mathcal{V}^2 \longrightarrow \mathbb{R}$$

is said to be *differentiable* at $A \in \mathcal{V}^2$ if there exists a second-order tensor $D\psi(A) \in \mathcal{V}^2$ such that

$$\psi(A + H) = \psi(A) + D\psi(A) : H + o(|H|), \quad \text{as } H \rightarrow O.$$

The tensor $D\psi(A)$ is called the *derivative* of ψ at A .



Proposition

If $\psi : \mathcal{V}^2 \rightarrow \mathbb{R}$ is differentiable at $A \in \mathcal{V}^2$, then its derivative is unique and

$$D\psi(A) : B = \left. \frac{d}{d\alpha} \psi(A + \alpha B) \right|_{\alpha=0}$$

for every $B \in \mathcal{V}^2$.

Proof.

See any good book on advanced calculus.





Result 2.21

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be an arbitrary frame. If

$$A = A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j,$$

then

$$D\psi(A) = \frac{\partial \psi}{\partial A_{ij}}(A) \mathbf{e}_i \otimes \mathbf{e}_j.$$



Proof.

Using the standard abbreviated notation, write

$$\psi(A) = \psi(A_{11}, A_{12}, \dots, A_{33}).$$

Let $\alpha \in \mathbb{R}$ be any scalar, and let

$$B = B_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$$

be an arbitrary second-order tensor. Then

$$\psi(A + \alpha B) = \psi(A_{11} + \alpha B_{11}, A_{12} + \alpha B_{12}, \dots, A_{33} + \alpha B_{33}).$$

By the chain rule,

$$\left. \frac{d}{d\alpha} \psi(A + \alpha B) \right|_{\alpha=0} = \frac{\partial \psi}{\partial A_{ij}}(A) B_{ij}.$$



Proof (Cont.)

By the preceding proposition,

$$\begin{aligned}
 D\psi(A) : B &= \left. \frac{d}{d\alpha} \psi(A + \alpha B) \right|_{\alpha=0} \\
 &= \frac{\partial \psi}{\partial A_{ij}}(A) B_{ij} \\
 &= \left(\frac{\partial \psi}{\partial A_{ij}}(A) \mathbf{e}_i \otimes \mathbf{e}_j \right) : (B_{k\ell} \mathbf{e}_k \otimes \mathbf{e}_\ell) \\
 &= \left(\frac{\partial \psi}{\partial A_{ij}}(A) \mathbf{e}_i \otimes \mathbf{e}_j \right) : B.
 \end{aligned}$$

Here i, j are dummy indices in the second line. In the third line, i, j, k, ℓ are dummy indices.



Proof (Cont.)

Since $B \in \mathcal{V}^2$ is arbitrary,

$$D\psi(A) = \frac{\partial \psi}{\partial A_{ij}}(A) \mathbf{e}_i \otimes \mathbf{e}_j.$$



Remark

The derivative of a scalar function $\psi : \mathcal{V}^2 \rightarrow \mathbb{R}$ is a second-order tensor-valued function

$$D\psi : \mathcal{V}^2 \longrightarrow \mathcal{V}^2.$$



Proposition

If $\psi : \mathcal{V}^2 \rightarrow \mathbb{R}$ is sufficiently smooth, then Taylor's theorem implies

$$\begin{aligned}\psi(A + H) &= \psi(A) + \frac{\partial \psi}{\partial A_{ij}}(A) H_{ij} + O(|H|^2) \\ &= \psi(A) + D\psi(A) : H + O(|H|^2), \quad \text{as } H \rightarrow O.\end{aligned}$$

Proof.

See any good book on advanced calculus. □



Remark

For a sufficiently smooth function, the preceding result shows that

$$\psi(A) + D\psi(A) : H$$

is a better approximation to $\psi(A + H)$ than the little-o remainder in Definition 2.20 alone indicates. In fact,

$$\psi(A + H) - \psi(A) - D\psi(A) : H = O(|H|^2).$$



Example

Consider the scalar function $\psi : \mathcal{V}^2 \rightarrow \mathbb{R}$ defined by

$$\psi(A) = \frac{1}{2} A : A = \frac{1}{2} A_{k\ell} A_{k\ell},$$

where $\mathcal{F} = \{\mathbf{e}_i\}$ is a frame and

$$A = A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j.$$

Then

$$\begin{aligned} \frac{\partial \psi}{\partial A_{ij}}(A) &= \frac{1}{2} \frac{\partial A_{k\ell}}{\partial A_{ij}} A_{k\ell} + \frac{1}{2} A_{k\ell} \frac{\partial A_{k\ell}}{\partial A_{ij}} \\ &= A_{k\ell} \frac{\partial A_{k\ell}}{\partial A_{ij}}. \end{aligned}$$

Here i, j are free, while k, ℓ are dummy.



Example (Cont.)

Since

$$\frac{\partial A_{k\ell}}{\partial A_{ij}} = \delta_{ik} \delta_{j\ell},$$

we obtain

$$\begin{aligned} \frac{\partial \psi}{\partial A_{ij}}(A) &= A_{k\ell} \delta_{ik} \delta_{j\ell} \\ &= A_{ij}. \end{aligned}$$

Therefore, by Result 2.21,

$$\begin{aligned} D\psi(A) &= A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j \\ &= A. \end{aligned}$$



Result 2.22

Let

$$\psi(S) = \det(S).$$

If $A \in \mathcal{V}^2$ is invertible, so that $\det(A) \neq 0$, then ψ is differentiable at A , and

$$D\psi(A) = \det(A)A^{-T}.$$



Proof.

Let $B \in \mathcal{V}^2$ be arbitrary. For $\alpha \neq 0$, set

$$\lambda = -\frac{1}{\alpha}.$$

Then

$$\begin{aligned}\psi(A + \alpha B) &= \det(A + \alpha B) \\ &= \det(\alpha A(A^{-1}B - \lambda I)) \\ &= \det(\alpha A) \det(A^{-1}B - \lambda I) \\ &= \alpha^3 \det(A) \det(A^{-1}B - \lambda I).\end{aligned}$$



Proof (Cont.)

Recall that the characteristic polynomial of a second-order tensor S is

$$\begin{aligned} p_S(\lambda) &= \det(S - \lambda I) \\ &= -\lambda^3 + I_1(S)\lambda^2 - I_2(S)\lambda + I_3(S), \end{aligned}$$

where

$$\begin{aligned} I_1(S) &= \text{tr}(S), \\ I_2(S) &= \frac{1}{2} [(\text{tr}(S))^2 - \text{tr}(S^2)], \\ I_3(S) &= \det(S). \end{aligned}$$

Apply this identity with $S = A^{-1}B$ and $\lambda = -1/\alpha$.



Proof (Cont.)

Substitution in the expression from the previous slide gives

$$\begin{aligned}\psi(A + \alpha B) &= \det(A) + \alpha \det(A) l_1(A^{-1}B) \\ &\quad + \alpha^2 \det(A) l_2(A^{-1}B) \\ &\quad + \alpha^3 \det(A) l_3(A^{-1}B).\end{aligned}$$

Both sides are polynomials in α . Since they agree for $\alpha \neq 0$, the identity also holds at $\alpha = 0$.

Therefore,

$$\left. \frac{d}{d\alpha} \psi(A + \alpha B) \right|_{\alpha=0} = \det(A) l_1(A^{-1}B).$$



Proof (Cont.)

Finally,

$$\begin{aligned} I_1(A^{-1}B) &= \text{tr}(A^{-1}B) \\ &= A^{-T} : B. \end{aligned}$$

Hence, for every $B \in \mathcal{V}^2$, the directional derivative is

$$\begin{aligned} \left. \frac{d}{d\alpha} \psi(A + \alpha B) \right|_{\alpha=0} \\ = \det(A) A^{-T} : B. \end{aligned}$$

Thus the candidate derivative is the linear map represented by

$$D\psi(A) = \det(A) A^{-T}.$$

It remains to verify the remainder required by Definition 2.20.



Proof (Cont.)

Taking $\alpha = 1$ and $B = H$ in the polynomial identity gives

$$\begin{aligned}\psi(A + H) - \psi(A) - \det(A)(A^{-T} : H) \\ = \det(A)l_2(A^{-1}H) + \det(A)l_3(A^{-1}H).\end{aligned}$$

Because A^{-1} is fixed, $l_2(A^{-1}H)$ is quadratic in H and $l_3(A^{-1}H)$ is cubic in H . Therefore,

$$\psi(A + H) - \psi(A) - \det(A)(A^{-T} : H) = O(|H|^2) = o(|H|).$$

Definition 2.20 now proves that ψ is differentiable at A and

$$D\psi(A) = \det(A)A^{-T}.$$





Exercise

Show that, for any $S, T \in \mathcal{V}^2$,

$$\text{tr}(ST) = S^T : T.$$

Proof.

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be a frame. Then

$$\begin{aligned}\text{tr}(ST) &= [ST]_{\ell\ell} \\ &= [S]_{\ell k} [T]_{k\ell} \\ &= [S^T]_{k\ell} [T]_{k\ell} \\ &= S^T : T.\end{aligned}$$

Here k, ℓ are dummy indices throughout. □



Result 2.23

Let S be a time-dependent second-order tensor, that is,

$$S : \mathbb{R} \longrightarrow \mathcal{V}^2.$$

If $S(t)$ is invertible, then

$$\begin{aligned} \frac{d}{dt} \det(S) &= \det(S) \operatorname{tr}(S^{-1} \dot{S}) \\ &= \det(S) S^{-T} : \dot{S}, \end{aligned}$$

where

$$\dot{S} := \frac{dS}{dt} := \frac{dS_{ij}}{dt} \mathbf{e}_i \otimes \mathbf{e}_j.$$



Proof.

Set $\psi(S) = \det(S)$ and use the abbreviated notation

$$\psi(S) = \psi(S_{11}, S_{12}, \dots, S_{33}).$$

By the chain rule and Result 2.21,

$$\begin{aligned} \frac{d}{dt} \psi(S) &= \frac{\partial \psi}{\partial S_{ij}}(S) \frac{dS_{ij}}{dt} \\ &= D\psi(S) : \dot{S}. \end{aligned}$$

Result 2.22 gives $D\psi(S) = \det(S)S^{-T}$. Therefore,

$$\begin{aligned} \frac{d}{dt} \det(S) &= \det(S)S^{-T} : \dot{S} \\ &= \det(S) \operatorname{tr}(S^{-1} \dot{S}). \end{aligned}$$





Tensor-Valued Functions



Definition (2.24)

A function

$$T : \mathcal{V}^2 \longrightarrow \mathcal{V}^2$$

is said to be *differentiable* at $A \in \mathcal{V}^2$ if there exists a fourth-order tensor $\mathbf{D}T(A) \in \mathcal{V}^4$ such that

$$T(A + H) = T(A) + \mathbf{D}T(A)H + o(|H|), \quad \text{as } H \rightarrow O.$$

The fourth-order tensor $\mathbf{D}T(A)$ is called the *derivative* of T at A .



Proposition

If $T : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ is differentiable at $A \in \mathcal{V}^2$, then its derivative is unique and

$$\mathbf{D}T(A)B = \left. \frac{d}{d\alpha} T(A + \alpha B) \right|_{\alpha=0}$$

for every $B \in \mathcal{V}^2$.

Proof.

See any good book on advanced calculus.





Result 2.25

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be an arbitrary frame, and suppose

$$T(A) = T_{ij}(A) \mathbf{e}_i \otimes \mathbf{e}_j.$$

If T is differentiable at $A \in \mathcal{V}^2$, then

$$\mathbf{D}T(A) = \frac{\partial T_{ij}}{\partial A_{k\ell}}(A) \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_\ell,$$

where

$$A = A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j.$$



Proof.

Use the abbreviated notation

$$T_{ij}(A) = T_{ij}(A_{11}, A_{12}, \dots, A_{33}).$$

For any $\alpha \in \mathbb{R}$ and

$$B = B_{k\ell} \mathbf{e}_k \otimes \mathbf{e}_\ell \in \mathcal{V}^2,$$

the chain rule gives

$$\left. \frac{d}{d\alpha} T_{ij}(A + \alpha B) \right|_{\alpha=0} = \frac{\partial T_{ij}}{\partial A_{k\ell}}(A) B_{k\ell}.$$

Here i, j are free component indices, while k, ℓ are dummy.



Proof (Cont.)

By the preceding proposition,

$$\begin{aligned}
 \mathbf{D}T(A)B &= \left. \frac{d}{d\alpha} T(A + \alpha B) \right|_{\alpha=0} \\
 &= \left. \frac{d}{d\alpha} (T_{ij}(A + \alpha B) \mathbf{e}_i \otimes \mathbf{e}_j) \right|_{\alpha=0} \\
 &= \left(\left. \frac{d}{d\alpha} T_{ij}(A + \alpha B) \right|_{\alpha=0} \right) \mathbf{e}_i \otimes \mathbf{e}_j \\
 &= \frac{\partial T_{ij}}{\partial A_{k\ell}}(A) B_{k\ell} \mathbf{e}_i \otimes \mathbf{e}_j.
 \end{aligned}$$



Proof (Cont.)

Recall that, if $\mathbf{C} \in \mathcal{V}^4$ and $U = \mathbf{C}B$, then

$$\begin{aligned} U &= (C_{ijkl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l) B \\ &= C_{ijkl} B_{kl} \mathbf{e}_i \otimes \mathbf{e}_j, \end{aligned}$$

where

$$B = B_{kl} \mathbf{e}_k \otimes \mathbf{e}_l.$$

Thus

$$\begin{aligned} \mathbf{D}T(A)B &= \frac{\partial T_{ij}}{\partial A_{kl}}(A) B_{kl} \mathbf{e}_i \otimes \mathbf{e}_j \\ &= \left(\frac{\partial T_{ij}}{\partial A_{kl}}(A) \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \right) B. \end{aligned}$$



Proof (Cont.)

Since $B \in \mathcal{V}^2$ is arbitrary,

$$\mathbf{D}T(A) = \frac{\partial T_{ij}}{\partial A_{k\ell}}(A) \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_\ell.$$



Remark

The derivative of a second-order tensor-valued function $T : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ is a fourth-order tensor-valued function

$$\mathbf{D}T : \mathcal{V}^2 \longrightarrow \mathcal{V}^4.$$



Proposition

If $T : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ is sufficiently smooth, then Taylor's theorem implies

$$T(A + H) = T(A) + \mathbf{D}T(A)H + O(|H|^2), \quad \text{as } H \rightarrow O.$$



Example

Suppose that $T : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ is defined by

$$T(A) = \text{tr}(A)A.$$

Then

$$T_{ij}(A) = \text{tr}(A)A_{ij} = A_{mm}A_{ij}.$$

The components of the derivative fourth-order tensor are

$$\begin{aligned} [\mathbf{D} T(A)]_{ijk\ell} &= \frac{\partial T_{ij}}{\partial A_{k\ell}}(A) \\ &= \frac{\partial}{\partial A_{k\ell}} (A_{mm}A_{ij}). \end{aligned}$$

Here i, j, k, ℓ are free, and m is dummy.



Example (Cont.)

Using the scalar product rule,

$$\begin{aligned}
 [\mathbf{D} T(A)]_{ijkl} &= \frac{\partial A_{mm}}{\partial A_{kl}} A_{ij} + A_{mm} \frac{\partial A_{ij}}{\partial A_{kl}} \\
 &= \delta_{km} \delta_{lm} A_{ij} + A_{mm} \delta_{ik} \delta_{jl} \\
 &= \delta_{kl} A_{ij} + A_{mm} \delta_{ik} \delta_{jl}.
 \end{aligned}$$

Thus

$$[\mathbf{D} T(A)]_{ijkl} = \delta_{kl} A_{ij} + A_{mm} \delta_{ik} \delta_{jl}.$$



Example (Cont.)

For any $B \in \mathcal{V}^2$,

$$\begin{aligned} \mathbf{D}T(A)B &= (\delta_{k\ell}A_{ij} + A_{mm}\delta_{ik}\delta_{j\ell}) B_{k\ell} \mathbf{e}_i \otimes \mathbf{e}_j \\ &= (B_{kk}A_{ij} + A_{mm}B_{ij}) \mathbf{e}_i \otimes \mathbf{e}_j \\ &= \text{tr}(B)A + \text{tr}(A)B. \end{aligned}$$

All indices in the tensor expansions are dummy indices. In the component identity, i, j are free and the contracted indices are dummy.



Examples



Example (Exercise 2.15)

Suppose that

$$\psi(A) = \text{tr}(A)(A : B),$$

where $B \in \mathcal{V}^2$ is a constant second-order tensor. Show that

$$D\psi(A) = (A : B)I + \text{tr}(A)B.$$

By Result 2.21,

$$D\psi(A) = \frac{\partial \psi}{\partial A_{ij}}(A) \mathbf{e}_i \otimes \mathbf{e}_j,$$

where $\mathcal{F} = \{\mathbf{e}_i\}$ is a frame and

$$A = A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j.$$



Example (Exercise 2.15, Cont.)

In components,

$$\psi(A) = A_{mm}A_{k\ell}B_{k\ell}.$$

Therefore,

$$\begin{aligned}\frac{\partial \psi}{\partial A_{ij}}(A) &= \frac{\partial}{\partial A_{ij}} (A_{mm}A_{k\ell}B_{k\ell}) \\ &= B_{k\ell} \left(\frac{\partial A_{mm}}{\partial A_{ij}} A_{k\ell} + A_{mm} \frac{\partial A_{k\ell}}{\partial A_{ij}} \right) \\ &= B_{k\ell} (\delta_{im}\delta_{jm}A_{k\ell} + A_{mm}\delta_{ik}\delta_{jl}).\end{aligned}$$

Here i, j are free, while k, ℓ, m are dummy.



Example (Exercise 2.15, Cont.)

Contracting the Kronecker deltas gives

$$\begin{aligned}\frac{\partial \psi}{\partial A_{ij}}(A) &= B_{k\ell} (\delta_{ij} A_{k\ell} + A_{mm} \delta_{ik} \delta_{j\ell}) \\ &= (A_{k\ell} B_{k\ell}) \delta_{ij} + A_{mm} B_{ij}.\end{aligned}$$

Hence

$$\begin{aligned}D\psi(A) &= [(A_{k\ell} B_{k\ell}) \delta_{ij} + A_{mm} B_{ij}] \mathbf{e}_i \otimes \mathbf{e}_j \\ &= (A : B) I + \text{tr}(A) B.\end{aligned}$$



Example (Derivative of $T(A) = A^2$)

Define $T : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ by

$$T(A) = A^2.$$

Since T is polynomial in the components of A , it is differentiable. For any $B \in \mathcal{V}^2$,

$$\begin{aligned} T(A + \alpha B) &= (A + \alpha B)^2 \\ &= A^2 + \alpha(AB + BA) + \alpha^2 B^2. \end{aligned}$$

Therefore, the directional characterization gives

$$\begin{aligned} \mathbf{D}T(A)B &= \left. \frac{d}{d\alpha} T(A + \alpha B) \right|_{\alpha=0} \\ &= AB + BA. \end{aligned}$$



Example (Derivative of $T(A) = A^2$, Cont.)

We verify the result in components. Since

$$T_{ij}(A) = A_{ik}A_{kj},$$

Result 2.25 gives

$$\begin{aligned} [\mathbf{D} T(A)]_{ijmn} &= \frac{\partial T_{ij}}{\partial A_{mn}}(A) \\ &= \frac{\partial A_{ik}}{\partial A_{mn}} A_{kj} + A_{ik} \frac{\partial A_{kj}}{\partial A_{mn}} \\ &= \delta_{im} \delta_{kn} A_{kj} + A_{ik} \delta_{km} \delta_{jn} \\ &= \delta_{im} A_{nj} + A_{im} \delta_{jn}. \end{aligned}$$

Here i, j, m, n are free indices, while k is dummy.



Example (Derivative of $T(A) = A^2$, Cont.)

Applying the fourth-order derivative to B gives

$$\begin{aligned}\mathbf{D}T(A)B &= (\delta_{im}A_{nj} + A_{im}\delta_{jn}) B_{mn}\mathbf{e}_i \otimes \mathbf{e}_j \\ &= (A_{nj}B_{in} + A_{im}B_{mj})\mathbf{e}_i \otimes \mathbf{e}_j \\ &= BA + AB.\end{aligned}$$

All indices are dummy in the tensor expansions. In the component identity, i, j are free, while m, n are dummy.

The component calculation agrees with the directional calculation.

$$\mathbf{D}T(A)B = AB + BA.$$